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MILITARY SPECIFICATION

TRACTOR, FULL-TRACKED, LOW-SPEED:

DIESEL-ENGINE-DRIVEN-MEDIUM-DRAWBAR PULL

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel-engine-driven, crawler type, medium-drawbar pull tractor.

1.2 Classification. The tractors shall be of the following types as specified (see 6.2):

Type I - Tractor equipped with Roll Over Protective Structure, or Semi-U or V (straight) type.

Type II - Tractor equipped with Roll Over Protective Structure, Semi-U or (straight) Tilt type blade and rear mounted towing winch.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards and handbooks. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2410

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SPECIFICATIONS

MILITARY

MIL-V-173	- Varnish, Moisture and Fungus Resistant (for Treatment of Communications Electronic and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-T-3351	- Tractor, Full-Tracked, Low-Speed; Tractor, Wheeled, Agricultural; and Tractor, Wheeled, Industrial; and Their Attachments, Packaging of.
MIL-M-7866	- Molybdenum Disulfide, Technical, Lubrication Grade.
MIL-E-10062	- Engines; Preparation for Shipment and Storage of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-I-14511	- Insulation Sheet, Cellular, Plastic, Thermal.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant, for Camouflage.
MIL-T-62314	- Test Equipment (Simplified) for Internal Combustion Engine (STE/ICE).

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-209	- Slinging Eyes and Attachments for Lifting and Tying Down Military Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS52131	- Connector, Plug, Electrical Intervehicle Power Cable.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

TACOM

12258941	- STE/ICE Diagnostic Connector.

REPORT

CR 82-588-003	- STE/ICE Design Guide for Vehicle Diagnostic Connector.

(Application for copies should be addressed to Commander, US Army Tank-Automotive Command, ATTN: DRSTA-RGD, Warren, MI 48090).

HANDBOOK

MILITARY

Air Force System Design Handbook DH 1-11	- Air Transportability.

(Copies of military specifications and military standards required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. Copies of military handbooks should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto.

TECHNICAL INFORMATION PACKAGE (TIP)

(Application for copies should be addressed to Belvoir Research and Development Center, ATTN: STRBE-NCE, Fort Belvoir, VA 22060.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN WELDING SOCIETY (AWS)

AWS D14.3 - Earth Moving and Construction Equipment. (Welding - Section 5 - Qualifications).

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(Application for copies should be addressed to the American Welding Society, 2501 Northwest 7th Street, Miami, FL 33125.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The tractors shall be diesel engine driven, crawler type, equipped with oscillating tracts and a hydraulic system for operation of all attachments specified in 3.16. The type I tractor shall be equipped with a roll over protective structure (ROPS), a semi-U (straight) type blade with hydraulic tilt and a 3-shank ripper bar equipped with 3 ripper shanks. The type II tractor shall be equipped with a roll over protective structure (ROPS), a semi-U (straight) type blade with hydraulic tilt and a rear mounted winch. The tractors shall be equipped with all components necessary to enable the tractor to function reliably and efficiently in all operations specified herin. The tractor shall conform to all Federal laws and regulations governing safety, noise levels, and pollution, which are in effect for this type of equipment at the time of issuance of the invitation for bids or request for proposal.

3.2 First article (preproduction model). The contractor shall furnish one or more of each type I and II tractors as specified (see 6.2) for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce tractors that comply with the requirements of this specification. Examinations and tests shall be as specified in section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.2). When specified (see 6.2), the Government will conduct any or all of the preproduction examinations and tests, as specified. Any changes or deviations of production tractors from the approved first article model(s) during production will be subject to the approval of the contracting officer. Approval of the first article model(s) will not relieve the contractor of his obligation to furnish tractors conforming to this specification.

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more production tractors for examination and tests as specified in 4.4.

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3.4 Material. All material used in the manufacture of the tractors shall be of the same type and quality as that being used by the contractor in his comparable production tractors for commerical market sales. Material shall be free from any defects or imperfections that would affect the tractor's operations or serviceability (see 6.4).

3.5 Safety. All rotating, reciprocating, or moving parts shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment, including exhaust pipes and other lines which may be subject to high temperatures or exposed to contact by personnel, shall be fully guarded or insulated. All platforms and steps shall have anti-skid surfaces. Grab irons shall be provided for mounting and dismounting from the tractor. Engine exhaust discharge shall be so directed that it does not endanger personnel.

3.5.1 Sound level. The tractor shall be equipped with all available standard and optional sound suppression equipment. If the sound level in the area occupied by the operator exceeds 85 dB(A) when measured in accordance with SAE J919, the contractor shall provide a caution plate and appropriate warning in the technical manual in accordance with MIL-STD-1474. The caution plate shall state: "CAUTION, HEARING PROTECTION REQUIRED WHEN OPERATING THIS MACHINE". If the sound level of cab equipped tractors with cab door(s) open is 85 dB(A), and when it is less than 85 dB(A) with the door(s) closed, a caution plate shall be provided stating: "CAUTION, HEARING PROTECTION REQUIRED WHILE OPERATING WITH DOORS OPEN." The caution plates shall have a yellow background with black lettering and shall be made of corrosion resistant material and permanently affixed to the tractor in a conspicuous, protected location in the operator's compartment. The exterior sound level of the tractor shall not exceed 88 dB(A) when measured in accordance with SAE J88.

3.5.2 Roll Over Protective Structure (ROPS) and Falling Objects Protective Structure (FOPS). Unless a cab in accordance with 3.5.3 or 3.5.4 is specified, the tractors shall be equipped with a bolted down or pin connected ROPS/FOPS canopy. The structure shall be removable and replaceable without cutting or welding. The ROPS/FOPS canopy shall conform to SAE J1040 and SAE J231, respectively. A seat belt conforming to SAE J386 shall be properly installed for use by the operator. Quick disconnect fittings (electric, etc.) shall be provided as applicable to facilitate ROPS/FOPS canopy removal. Two or more slinging eyes shall be provided. Each eye shall be capable of supporting the total weight of the ROPS/FOPS device. The slinging eyes shall be positioned to allow suspension of the ROPS/FOPS in its normal attitude. The rear of the ROPS/FOPS canopy shall be covered by a protective screen. A rear view mirror shall be provided. Front sweeps are not required.

3.5.3 Cab. When specified (see 6.2), a fully enclosed, bolted down or pin connected cab shall be provided in lieu of the ROPS/FOPS canopy specified in 3.5.2. The cab shall be protected by an integral or external ROPS/FOPS structure conforming to the requirements of 3.5.2. A seat belt conforming to SAE J386 shall be properly installed for use by the operator. The cab shall be removable and replaceable without cutting or welding. The cab shall be provided with a minimum of four slinging eyes. Each eye shall be capable of supporting

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the full weight of the cab. Quick disconnect fittings (electric, etc.) shall be provided as applicable to facilitate cab removal. The cab shall be equipped with a minimum of one door and another emergency exit on the opposite side of the cab (either a door or window that can be opened), windows and front and rear windshields. The door(s) of the cab shall have a latch mechanism that will latch the door(s) in the full open position. The front and rear windshields shall be equipped with windshield wipers. The cab shall be equipped with a hot water personnel heater equipped with one or more shut-off valve(s) in the coolant lines to stop or vary the flow of coolant to the heater. The capacity of the heater shall be sufficient to raise the temperature inside the cab from minus 25° F to plus 35° F within one hour of heater start. Ventilation fans or defrosters shall be furnished that direct air flow to at least 75 percent of the front and rear windshields and side door(s) and side window(s). Each fan shall have an on-off switch and shall be directionally adjustable. A rear protective screen and a rear view mirrow shall be provided. Front sweeps are not required.

3.5.4 Winterized cab. When specified (see 6.2), a quantity of tractors as specified (see 6.2) shall be equipped with winterized cabs. In addition to meeting the requirements for the cab specified in 3.5.3, the winterized cab shall also meet the following requirements: Insulation for the cab shall comply with the requirements of MIL-I-14511, 1/2 inch. The adhesive for bonding the insulation shall retain its adhesive properties at ambient temperatures to minus 60° F. The capacity of the personnel heater shall be sufficient to raise the temperature inside the cab from minus 50° F to plus 35° F within one hour of heater start.

3.5.5 Horn. A conventional operator controlled horn shall be furnished. The horn may be hand or foot operated and shall be distinguishable from the surrounding noise level.

3.5.6 Backup alarm. A backup alarm conforming to SAE J994, type C or D shall be provided. The backup alarm signal shall be audible above, and distinguishable from the surrounding noise level.

3.5.7 Human factors. The size and location of access/egress passages shall be in accordance with SAE J185. Components and equipment removed, disassembled or relocated for operation or transportability as specified in 3.17 and 3.17.1, which must be lifted or carried by operating personnel shall not exceed 50 pounds in weight for one person lifts and 100 pounds for two or more person lifts and shall be provided with suitable handles or grasp areas.

3.6 Physical characteristics.

3.6.1 Weight limits. The type I and II tractors without the removable ROPS canopy/ROPS cab, blade (not including hydraulic lift cylinders), push beams, trunnions, rear mounted winch and ripper, but with all other acessories, coolant, lubricating and hydraulic systems filled to capacity, and with not less than 25 percent fuel, shall weigh not more than 39,600 pounds and not less than 34,000 pounds.

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3.6.2 Ground clearance. The ground clearance of the tractor, excluding grouser height, shall be not less than 12 inches measured along an area 15 inches to each side of the longitudinal centerline of the tractor.

3.7 Environmental conditions. Without use of any external means, equipped with only standard or optional equipment, the engine shall start within 5 minutes and the tractor shall operate under full load conditions within 15 minutes after engine start in any ambient temperature from minus 25° F to plus 120° F. Cold start aids must be integral with the tractor and no aids may be used which rely upon external support such as an electrical power source. Hand-held aerosol cans are not permitted. The tractor shall be capable of operating in hot, humid tropical environments and dry, dusty conditions. The tractor shall be capable of being stored for extended periods in temperatures ranging from minus 28° F to plus 160° F without damage.

3.7.1 Fungus and moisture resistance. The electrical circuitry, including all components and connections, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.7.2 with 1 percent copper 8-quinolinolate (by weight) or 7 percent salicylanilide based on the nonvolatile content of the varnish except:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will ne adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both, to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.7.2 Composition. Composition I shall be used unless prohibited by local air pollution regulations governing the application of the varnish. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management District.

3.8 Performance.

3.8.1 Capability. The type I tractor shall be capable of performing all dozing operations performed with a semi-U (straight) type tilt blade and ripping to a depth of not less than 24 inches. The type II tractor shall be capable of performing all dozing operations performed with a semi-U (straight) type tilt blade and shall have a winching capacity of at least 50,000 pounds at a line speed of not less than 80 feet per minute.

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3.8.2 Stability. The type II tractor shall operate on 35 percent side slopes and 60 percent longitudinal slopes, or any combination thereof, without either track losing contact with the ground. All moving parts shall be lubricated when the tractor is operating under these conditions, with no loss of hydraulic power to attachments or tractor control system. The fuel system shall maintain operating pressure when operating on these slopes.

3.8.3 Fording. The tractors shall start and operate during and after shallow fording in fresh water to any depth up to and including 30 inches, including wave action, without damage or leakage of water into reservoirs containing lubricants.

3.8.4 Travel speeds. The tractors, equipped as specified herein, without towed load, shall travel at not less than the following speeds when the engine is operating at full-load governed speed:

a. Low gear forward - 2.0 mph.
b. High gear forward - 5.8 mph.
c. Low gear reverse - 2.3 mph.
d. High gear reverse - 6.5 mph.

3.8.5 Drawbar pull. The type II tractor, equipped with all attachments and accessories specified herein shall deliver not less than 35,000 pounds drawbar pull at a speed of not less than 1.4 mph. This pull shall be reserve tractive force as defined in SAE J872.

3.8.6 Drawbar pull vs speed. The drawbar pull of the tractor at speeds of 1.4 mph to not less than 4.0 mph shall be such that, when plotted, no point of the maximum resulting curve formed between the intersect points of the successive gear ratio curves shall fall below the curve shown in figure 1.

3.9 Components. The tractor and all of its components shall be used within currently published rated capacities. All power train components shall be certified by the contractor as being compatible with and properly matched with all related or affected components assembled to meet this specification.

3.10 Engine. For the purpose of selecting the engine, the tractor shall be classified as a class II end item in accordance with MIL-STD-1410. The engine shall be equipped with a cold starting aid system of the glow plug or ether primer type, conforming to MIL-STD-1410. The engine furnished shall be capable of operating on fuels and lubricating oils as specified in MIL-STD-1410. The engine shall be equipped with a muffler and an exhaust stack with a rain cap or be fabricated and mounted to prevent the entry of rain or snow.

3.10.1 Engine horsepower. The applied net horsepower of the engine at the flywheel when measured in accordance with SAE J816, shall be not less than 200 horsepower.

3.10.2 Fuel system. The fuel tank shall have sufficient capacity for 10 hours of continuous dozing operation as specified in 4.6.2.7. The fuel tank joints shall be welded. The fuel tank opening shall be provided with a fuel

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strainer. The fuel tank shall be provided with shut-off valves for all fuel lines from the tank. Fuel lines and fittings shall be located, routed or restrained in protected locations to prevent damage due to chafing or vibration. Electrical wiring shall not be attached to or in contact with the fuel lines. The fuel tank and fuel filter shall be provided with a means for draining water and sediment from the fuel system. The mounting of the fuel tank shall prevent shifting, damage and fatigue as a result of travel and transport.

3.10.3 Oil sampling valve. An oil sampling valve shall be provided on the engine and transmission. The valve shall be manually operated and shall automatically close when released. It shall be made of material resistant to corrosion such that it will not contaminate the sample. The discharge port of the valve shall be covered with a captive chained cap conforming to SAE J514. The valve shall be located in such a way as to insure accessibility for taking samples and that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running. Sampling valves shall be labeled crankcase or transmission, whichever is appropriate to indicate the type of oil sampled.

3.10.4 Tachometer drive. A heavy-duty key type tachometer drive shall be provided and shall have a 5/32-inch female connection in accordance with SAE J678. The output drive shall be covered with a threaded, captive chained cap, for protection when not in use.

3.10.5 Engine air cleaner. The engine shall be equipped with an air cleaner of the dual element type with a precleaner, automatic dust ejector and an air cleaner restriction indicator.

3.11 Power train. A power train consisting of a torque converter; a transmission; two individually controlled, power-assisted or power-operated steering devices; gear-type final drive units; and connecting components shall be furnished. The steering devices shall transmit the full power of the engine to either track. Brakes shall be provided which lock both tracks simultaneously and lock either track individually to permit pivot turning. The final drive units shall be enclosed in dustproof and waterproof housings.

3.11.1 Transmission. The transmission shall be the full power-shift type in combination with a torque converter. The transmission shall provide not less than three forward and three reverse gear ratios, all of which shall be manually selected. All transmission components and material shall be compatible with oils conforming to MIL-L-2104, grade 10 and MIL-L-46167. The transmission shall be provided with filter(s) having replaceable element(s) and sized in accordance with the transmission manufacturer's recommendation. The transmission control lever shall be mounted within reach of the operator when in his normal seated poisition. The shift lever shall pass through the neutral position when shifting from any forward to any reverse gear. A manual lock shall be provided to lock the transmission shift lever in the neutral position to prevent accidental transmission engagement during starting and periods of engine idling. A safety device shall be provided to prevent starting of the engine while the drive train is engaged.

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3.11.2 Brakes. Brakes shall be provided on the tractor which will prevent the track from turning when a force of 35,000 pounds is applied to either the drawbar or the towhook when the engine is shut down. Tractors that are equipped with brakes which are automatically applied upon engine shut-down shall be equipped with a means of manually releasing the brakes for towing or retrieval purposes.

3.11.3 Track mechanism. Track links and shoes shall be heat-treated steel of drop-forged or rolled construction, and all like parts shall be interchangeable. Track links shall be equipped with sealed and lubricated track pins. The track chains shall have a quick-disconnect master link of the split link or similar type. Recoil springs shall be provided for cushioning of the front idlers. The tracks shall be equipped with hydraulic track adjusters. The adjusters shall provide a means for filling and a manually operated bleeder valve. The track adjusters shall be located in a protected location to prevent damage by rocks or dirt.

3.11.4 Track oscillation. The tractors shall have a total vertical track oscillation of not less than 10 inches and not more than 16 inches measured at the center of the front idler.

3.11.5 Track shoes. The tractors shall be equipped with clipped-corner, full-grouser type shoes. Shoes shall be not less than 20 inches wide.

3.11.6 Track gage. The track gage shall be not less than 78 inches.

3.11.7 Track roller guards. Inner and outer track roller guards shall be furnished for each track assembly. The guards shall provide continuous coverage for the full length of the track frame.

3.11.8 Sprocket guards. When spoke or web-type sprockets are furnished, each sprocket shall be protected by guards completely covering the open portion.

3.11.9 Front idler guards. When spoke or web-type front idlers are provided, inner and outer guards shall be furnished to completely cover the open portion of each front idler above the track frame.

3.12 Accessories.

3.12.1 Rock guard. The underside of each tractor shall be provided with rock guards to withstand, without damage that could affect the serviceability of the tractor, blows from rocks, tree stumps, and similar obstacles. The rock guard shall protect the engine, all drive-line components, and transmission housings and shall be removable. The rock guard shall be sectionalized to facilitate maintenance and shall be equipped with the necessary service holes for maintenance. Each service hole more than 16 square inches in size shall be equipped with a removable cover plate of the same material and thickness as the rock guard.

3.12.2 Radiator guard. A front mounted radiator guard shall be provided. The guard shall be hinged to facilitate cleaning of the radiator.

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3.12.3 Front tow hook. A tow hook shall be provided on the front of the tractors. The throat opening of the tow hook shall be not less than 1-1/2 inches. The tow hook and its mounting shall be capable of withstanding a 75,000 pound pull.

3.12.4 Equalizer spring. An equalizer spring or equalizer bar shall be furnished.

3.13 Operator's compartment.

3.13.1 Seat. An adjustable commercial operator's seat shall be provided. The seat shall be located to provide the operator with convenient access to operating controls, and visibility of the work zone and all instruments and gages mounted in the operator's compartment when seated with seat belt fastened and properly adjusted.

3.13.2 Instruments and indicators. The following instruments and indicators shall be provided and shall be located on the instrument panel except the hourmeter, the hydraulic oil level indicator, the fuel level indicator and the air cleaner restriction indicator, which may be remotely mounted:

a. Engine lubricating oil pressure gage.
b. Engine cooling liquid temperature gage.
c. Hourmeter.
d. Transmission/torque converter oil temperature or pressure gage or warning light.
e. Ammeter, voltmeter, or battery charging indicator.
f. Hydraulic oil level indicator (dipstick, sight tube or gage).
g. Fuel level indicator (dipstick or gage).

3.13.3 Controls. All controls for the tractors and their attachments shall be located in the operator's compartment and shall be within reach of the operator in his normal operating position. All controls shall be identified by decals or plates on or adjacent to each control to indicate the function or use.

3.13.4 Manual engine control. The manual engine control shall be hand operated and designed to secure the throttle linkage in any setting throughout the entire range of adjustment from idle speed to maximum rated rpm.

3.13.5 Decelerator. A foot operated decelerator shall be provided. The decelerator shall be accessible to the operator's foot when he is in the normal seated position. Pushing on the decelerator shall overide the manual engine speed control and shall decrease the engine speed from the throttle setting to approximately low engine idle. Releasing the decelerator shall return the engine speed to the control and setting of the throttle.

3.13.6 Dozer control lever. The tractors shall be provided with a single dozer control lever which shall be capable of raising, lowering, holding, tilting and floating the dozer.

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3.14 Hydraulic system. The hydraulic system shall provide control of all hydraulic functions for all operations specified herein. Hydraulic filtration shall be in accordance with SAE J931. The system shall provide sufficient force and means for operation of the dozer, ripper and the winch, when operated as specified herein. The system shall be compatible with oil conforming to MIL-L-2104, grade 10 except that for low temperature operation, oil conforming to MIL-L-46167 may be used. Hydraulic hose shall conform to SAE J517, 100R1, 100R2, or 100R10 for pressure lines and 100R4 for suction lines. Tubing shall be in accordance with SAE J524 or J525, or be AISI 4130 annealed steel. Diagnostic ports for measuring temperature and pressure shall be provided in accordance with SAE J1298.

3.15 Electric system. The tractors shall be equipped with a 24-volt electrical system.

3.15.1 Batteries. Commercial batteries shall be provided. The battery and cable terminals shall be identified with their respective polarity by raised or recessed markings. Corrosion resistant nuts and bolts shall be provided. The batteries shall be readily accessible for servicing.

3.15.2 Battery enclosure. A weather protected box or compartment type enclosure shall be provided. The enclosure shall include a drain for water or electrolyte. A vent to prevent gas accumulation shall be provided. The enclosure shall be equipped with battery holddown components to prevent damage to the batteries. The enclosure shall be treated to prevent corrosion.

3.15.3 Alternator. A commerical alternator with an output of not less than 50 amps shall be provided. The alternator shall be of the brushless type with integral regulator and shall be protected from reverse polarity.

3.15.4 Lighting. The following minimum number of lights with associated circuits, circuit-breakers, or other protective devices and switches shall be provided: two headlights, one rear light and one dash light. The front and rear lights shall be guarded or mounted in a protected location.

3.15.5 Electromagnetic radiation. The tractors shall comply with the electromagnetic radiation limits of SAE J551. Suppressors, when used, shall conform to SAE J552.

3.15.6 Slaving Components. The tractor shall be equipped with a 24 volt slave receptacle conforming to MS52131. The slave receptacle shall permit charging of the batteries and slave starting of the engine from an external power source and shall also provide a power source for charging and slaving other equipment. The slave receptacle shall be installed on the exterior of the tractor near the battery compartment and shall be accessible to personnel standing on the ground. The voltage potential of the slave receptacle shall be identified on a plate mounted adjacent to the receptacle. The plate shall be in accordance with the requirements of 3.24.

3.15.7 Starter disconnect. The cranking system components shall be protected by an automatic starter disconnect system. The system shall automatically

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disconnect the starter or starter solenoid when the engine starts, and shall prevent starter engagement when the starter switch is activiated while the engine is running.

3.15.8 Diagnostic connector assembly (DCA). The vehicle shall incorporate an easily accessible DCA for Simplified Test Equipment/Internal Combustion Engine (STE/ICE) in accordance with Vehicle Diagnostic Connector Assemblies, Report No. CR-82-588-003 and MIL-T-62314, Appendix B. The diagnostic connector, TACOM drawing 12258941, on the vehicle will interface with the STE/ICE test equipment. As a minimum, the DCA shall be capable of performing the functions defined in table 2 of Report No. CR-82-588-003. A separate wiring harness shall include all wiring and necessary hardware to perform required capabilities.

3.16 Attachments.

3.16.1 Bulldozer. The dozer shall be a semi-U (straight) type and shall be hydraulically operated and shall consist of a moldboard with any standard or optional liners, a push plate, cutting edges, end bits, push beams, hydraulic tilt cylinder and an adjustable strut. The push plate shall be not less than 30 inches high, 41 inches wide and 1/2 inch thick, constructed to fit the curvature of the moldboard and fully welded to the moldboard.

3.16.2 Winch. The winch shall be a single drum, reversible, hydraulically controlled, overwind towing winch. Winch controls shall be located in the operator's compartment. One hundred seventy-five feet of 1-inch wire rope with a hook shall be furnished on the winch. The hook's capacity shall be greater than the breaking strength of the wire rope. The winch shall have a bare drum line pull of not less than 50,000 pounds at a speed of not less than 80 feet/minute. If the winch must be removed to meet transportability requirements specified herein, a gasket sealing cover plate for the winch drive opening and winch drive pump opening (if required) shall be provided.

3.16.3 Ripper. A radial or parallelogram type ripper shall be provided on the type I tractor. The ripper shall consist of a mounting bracket, three-shank carrier beam, three straight shanks, shank protectors and hydraulic components necessary for operation. Each shank shall have two pin holes for adjustment and shall have replaceable points. The ripper shall have a digging depth of not less than 24 inches. The ripper shall have sufficient hydraulic force to lift the rear of the tractor clear of the ground and raise the ripper from maximum digging depth to the normal carry position. The ripper, in travel position, shall not be higher than the tractor and shall maintain the overall ground clearance specified in 3.6.2 when raised to maximum height. Ripper controls shall be located in the operator's compartment of the tractor.

3.16.4 Drawbar or towing coupler. A drawbar or towing coupler integral with the winch specified in 3.16.2 shall be furnished. A drawbar or towing coupler pin meeting the requirements of SAE J749, for a 60,000 pound tractor shall be provided. The pin shall be provided with a locking device permanently attached to the drawbar or towing coupler.

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3.17 Transportability. The tractors shall be capable of being transported by truck trailers, rail, ocean going carriers, and C-130 and C5A aircraft and withstand the impact forces encountered in shipment without damage or permanent deformation. Complete diagrams and instructions for lifting and tying down the tractors on freight cars and truck trailers shall be furnished in the operators' manual. The operators' manual shall also include instructions for removal of attachments or components when required for transport by above modes.

3.17.1 Air tranportability. The tractors shall be capable of being loaded (drive on) and transported by the C-130 and C5A aircraft. The maximum reduced weight of the tractors for C-130 transport shall not exceed 39,600 pounds, 3,000 pounds/linear foot of each track, or a bearing pressure of 50 psi. The width of the tractor shall not exceed 105 inches. The overall reducible height of the tractor, including grousers, shall not exceed 100 inches. Additional data including cargo envelopes and allowable floor loadings are found in AFSC Design Handbook DH1-11. Disassembly to achieve air transportable configuration of ROPS canopy equipped units shall be accomplished in not more than 6 clock hours with unlimited manpower and reassembly shall be accomplished in not more than 6 clock hours with unlimited manpower using only common tools and available lifting devices. An additional 4 clock hours for disassembly and 4 clock hours for resassembly, with unlimited manpower, is allowed for cab equipped units. The tractors shall be capable of being driven on and off the aircraft in the transportable configuration. The tractor's fuel tank shall contain a minimum of 1/4 its fuel capacity for air transport.

3.18 Slinging provisions. The tractors shall be provided with four slinging provisions conforming to MIL-STD-209, class 1 or 3, type II. The provisions shall enable the tractors to be lifted in the normal traveling or operating position. Provisions shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provision without weld failure or permanent deformation of the tractor or the provision.

3.19 Tiedown provisions. The tractor shall be provided with integral provisions to permit tiedown of the tractor to the floor or deck of the transportation medium. The tiedown provisions shall comform to MIL-STD-209, class 2 or 3, type II and shall withstand stresses in the amount and direction of pull specified without weld failure or permanent deformation of the tractor or the provision.

3.20 Shipping data plates. Shipping data plates for each tractor shall be furnished and shall show the silhouette of the tractor indicating the location and capacity of slinging provisions, tiedown provisions, the center of balance and shipping weight. Plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class I material with black backgound and light lettering. The plates shall be attached to the left side of the tractor and shall show the location as viewed from the left side.

3.21 Vehicle classification data. The contractor shall complete and submit the vehicle classification data sheets, figures 2 and 3, within the time frame specified (see 6.2).

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3.22 Stenciling instructions. Stenciling instructions shall be provided with each set of technical publications. The instructions shall be provided in card form showing location and wording for tiedown and slinging provisions and the center of gravity. Each tractor shall be stenciled using the instructions provided with the technical publications. Stenciling of US Army Registration numbers and other markings shall be applied in accordance with MIL-STD-642. Registration numbers shall be furnished by the contracting officer.

3.23 Identification plates and marking. Each tractor shall be furnished with an identification plate showing the contractor's model number, NSN, USA registration number, serial number, weight, date of manufacture and date of delivery. The identification plate shall be permanently affixed to the tractors in a readily accessible and conspicuous location. Plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class I material with black background and light lettering. All attachments or components removed or disassembled for shipment shall be matchmarked for proper reassembly.

3.24 Instruction plates. The tractors shall be equipped with instruction plates including warnings and cautions suitably located and permanently affixed, describing any special or important procedure to be followed in operating and servicing the tractor. Plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class I material with black background and light lettering.

3.25 Treatment and painting. The interior and exterior portions of the tractor normally painted shall be cleaned, treated, primed and painted in accordance with MIL-T-704, type F and the finishing coat shall be in accordance with MIL-C-46168, camouflage green 383 color. MIL-C-46168 finish coat may be applied directly over existing alkyd paint without additional surface preparation except for cleaning, if the base alkyd paint meets applicable military/Federal specifications and if the base alkyd paint is allowed to cure for a period of 14 days.

3.26 Lubrication. A lubrication means shall be provided for all moving parts requiring lubrication. Each tractor shall be serviced and run-in with military lubricants conforming to MIL-L-2104, MIL-L-2105, MIL-L-46167, MIL-M-7866, MIL-G-10924, and MIL-G-23827 as applicable. An instruction plate in accordance with 3.24 showing complete lubrication instructions, the applicable military lubricants and their temperature range shall be furnished.

3.27 Maintenance.

3.27.1 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners (except rock guards). Drain outlets shall be located for accessibility.

3.27.2 Filling, draining and checking provisions. The hydraulic system reservoir, the transmission housing, and all other enclosures which contain lubricant or hydraulic oil shall be equipped with dipsticks, sight gages, or with

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checkplugs conforming to SAE J371 to determine the fluid levels. Each enclosure shall be equipped for filling and for draining. Each drainplug or valve shall be located so that removal of the plug or valve will result in complete drainage of the fluid from the enclosure (except torque converter). When the tractor is in a level position, drainage of fluids shall be to a container below the tractor without draining on or over any part of the tractor. Integral tubes, hose, or troughs may be used to convey the fluids from the drain outlet. Accessibility to the fill opening, the fluid-level check device, and the drainplug or valve shall be provided without removal or adjustment of accessories or parts, except for plates equipped with hand-operable, quick disconnect fasteners. The radiator drain shall be such that coolant will not drain onto any part of the tractor. Extension hose is permitted.

3.27.3 Maintenance schedule. The contractor shall provide a maintenance schedule prior to the tests required by 4.6.2. The maintenance schedule shall identify the adjustments, lubrication and time intervals for performing the scheduled maintenance.

3.28 Technical publications. Such technical publications as are specified (see 6.2) shall be furnished.

3.29 Special tools. Any nonstandard or special tools required for equipment operation and operator maintenance of the tractors shall be furnished with the tractors.

3.30 Toolbox. A lockable weatherproof storage compartment(s) or toolbox of sufficient size to hold all tools and equipment required for operation and operator maintenance shall be provided.

3.31 Technical information package. The contractor shall complete and submit a technical information package within the time frame specified (see 6.2), depicting the tractor and its components exactly as it will be furnished.

3.32 Anti-vandalism protection. The tractor shall be equipped with locks to secure the following:

a. Fuel tank filler cap.
b. Hydraulic oil reservoir filler cap.
c. Transmission oil filler cap.
d. Engine oil filler cap.
e. Radiator filler cap.
f. Tool box.
g. Battery box.
h. Cab door(s) (when cab is supplied).

When any of the above items are secured by lockable doors or lockable engine side panels, individual locks are not required. Keys for ignition, cab doors, cover panels, filler caps and battery/tool boxes shall be interchangeable to the maximum extent possible. Two sets of keys shall be furnished with each tractor and each set shall be identical for all tractors.

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3.33 Government-furnished equipment (GFE). When specified (see 6.2), the following items shall be furnished by the Government (see 6.3):

a. Case, NSN: 7520-00-559-9618 complete with maintenance forms. (The contractor shall furnish the fasteners and install the case on the tractors.)
b. Vehicle weight classification sign set, NSN 9905-00-565-6267.
c. STE/ICE, NSN: 4910-00-124-2554, for test in 4.6.2.18.1.

3.34 Vehicle weight classification sign set. The contractor shall apply the vehicle weight classification number to each vehicle utilizing a vehicle weight classification sign set (see 3.33). The set shall be bolted to the front of the tractor in a location approved by the contracting officer. The contracting officer shall assign the classification numbers to be displayed on the tractor.

3.35 Workmanship.

3.35.1 Casting and forgings. All parts, components, and assemblies of the tractor which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework material shall meet typical mechancial properties of new casted or forged material and not affect fit or function.

3.35.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. All bends shall be made with controlled means to insure uniformity of size and shape. Precautions shall be taken to avoid overheating. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.35.3 Welding and welders. Welding procedures and welder qualifications shall be in accordance with AWS D14.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspections. The contractor is responsible for insuring that components and materials used are manufacturered, examined and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3).
b. Initial production inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Inspection comparison (see 4.7).
e. Inspection of packaging (see 4.8).

4.3 First article inspection. The first article (preproduction model) tractors (see 6.2), furnished by the contractor shall be examined (see 4.3.1.1 and 4.3.1.2), and tested (see 4.3.2), to determine compliance with the requirements of this specification. Prior to the examination and test of the tractors, the following shall be performed:

a. Service the tractors with the lubricants specified.
b. Break in the tractors as prescribed by the contractor.

4.3.1 Examinations.

4.3.1.1 Pretest examination. Prior to the first article tests, the type I and type II tractors shall be examined for the defects marked "X" in column 1 of table I. Presence of one or more defects shall be cause for rejection. Prior to the first article tests the tractors shall be examined and compared with this specification, with the contractor's published brochures, with the commercial manuals, and with the completed Technical Information Package. Any discrepancy between the specification, brochures, manuals, or Technical Information Package and the first article models shall be cause for rejection. Prior to the first article tests, certifications and certified test reports required by 4.6.2.21, and 4.6.2.22 shall be examined. Incomplete, inaccurate, or otherwise unacceptable certifications, or certified test reports shall be cause for rejection.

4.3.1.2 Post-test examination. Upon successful completion of all tests specified in 4.3.2, the type I and type II tractors shall be subjected to the examinations marked "X" in column 2 of table I. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1.1, the tractors shall be subjected to the tests marked "X" under column 1 of table II. Failure of any test, any evidence of leakage, damage, breakage, or deformation after any or all of the tests or an unacceptable test report shall be cause for rejection.

4.3.3 Test report. The contractor shall prepare and submit the first article test report to the contracting officer for approval within the time frame specifed (see 6.2). This report shall include a detailed description of all examinations and tests performed, deficiencies encountered, and actions taken to correct the deficiencies.

4.4 Initial production inspection. When specified (see 3.3) one or more type I and type II initial production tractors will be selected at random by the Government from the tractors being produced by production tooling and will be

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examined for the defects marked "X" in column 3 of table I and subjected to the tests marked "X" in column 2 of table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of the initial production tractors shall not exclude the remaining tractors from the quality conformance inspection and acceptance provisions specified in section 4. In addition to any test specified as part of the inital production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test. Failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

TABLE I. Examination schedule.

First Article(s)		Initial	Quality		Requirement
Pre-Test	Post Test	Production	Conformance	Defects	Paragraph(s)
1	2	3	4	5	6
X	-	X	X	101. Assembly not complete, components missing, or not as specified.	3.1
X	X	X	X	102. Material not as specified or rust, corrosion or deterioration has developed.	3.4
X	X	X	-	103. Safety characteristics not as specified.	3.5
X	-	X	X	104. Caution plate missing (if required).	3.5.1
X	-	X	-	105. ROPS/FOPS not as specified.	3.5.2
X	-	X	-	106. Winterized cab not as specified (when required).	3.5.4
X	-	X	-	107. Horn not as specified.	3.5.5
X	-	X	-	108. Backup alarm not as specified.	3.5.6
X	X	X	-	109. Human factors not as specified.	3.5.7
X	-	X	-	110. Ground clearance not as specified.	3.6.2
X	-	X	X	111. Fungus and moisture resistance not as specified.	3.7.1
X	-	X	X	112. Engine not as specified, does not function.	3.10
X	-	X	X	113. Exhaust system not as specified.	3.10
X	-	X	X	114. Fuel system not as specified.	3.10.2

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TABLE I. Examination schedule. (Cont'd)

First Article(s)		Initial	Quality	Defects	Requirement
Pre-Test	Post Test	Production	Conformance		Paragraph(s)
1	2	3	4	5	6
X	-	X	X	115. Oil sampling valve not as specified.	3.10.3
X	-	X	X	116. Tachometer drive not as specified.	3.10.4
X	-	X	X	117. Engine air cleaner not as specified.	3.10.5
X	-	X	X	118. Power train not as specified.	3.11
X	-	X	X	119. Transmission not as specified.	3.11.1
X	-	X	X	120. Brakes not as specified.	3.11.2
X	-	X	X	121. Tracks not as speicifed.	3.11.3
X	-	X	-	122. Track oscillation not as specified.	3.11.4
X	-	X	X	123. Track shoes not as specified.	3.11.5
X	-	X	-	124. Track gage not as specified.	3.11.6
X	-	X	X	125. Track roller guards not as specified.	3.11.7
X	-	X	X	126. Sprocket guards missing (if required).	3.11.8
X	-	X	X	127. Front idler guards missing (if required).	3.11.9
X	-	X	X	128. Rock guard not as specified or damaged.	3.12.1
X	-	X	X	129. Radiator guard not as specified.	3.12.2
X	-	X	X	130. Front tow hook missing.	3.12.3
X	-	X	X	131. Equalizer spring missing.	3.12.4
X	-	X	X	132. Seat not as specified.	3.13.1
X	X	X	X	133. Instruments and indicators not as specified or inoperable.	3.13.2
X	-	X	X	134. Controls not as specified.	3.13.3
X	-	X	X	135. Manual engine control not as specified.	3.13.4
X	-	X	X	136. Decelerator not as specified.	3.13.5
X	-	X	X	137. Dozer control not as specified.	3.13.6

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TABLE I. Examination schedule. (Cont'd)

First Article(s) Pre-Test	First Article(s) Post Test	Initial Production	Quality Conformance	Defects	Requirement Paragraph(s)
1	2	3	4	5	6
X	-	X	X	138. Hydraulic system not complete, missing components, does not function or not as specified.	3.14
X	X	X	X	139. Electrical system not complete, does not function, or not as specified.	3.15
X	-	X	X	140. Batteries not as specified.	3.15.1
X	-	X	X	141. Battery enclosure not as specified.	3.15.2
X	-	X	X	142. Alternator not as specified.	3.15.3
X	-	X	X	143. Lighting not as specified.	3.15.4
X	-	X	X	144. Slaving components not as specified.	3.15.6
X	-	X	X	145. Starter disconnect not as specified.	3.15.7
X	-	X	X	146. DCA not as specified.	3.15.8
X	-	X	X	147. Bulldozer not as specified.	3.16.1
X	-	X	X	148. Winch not as specified.	3.16.2
X	-	X	X	149. Ripper not as specified.	3.16.3
X	-	X	X	150. Drawbar or towing coupler not as specifed.	3.16.4
X	-	X	X	151. Slinging provisions damaged, missing, incorrectly located or not as specified.	3.18
X	-	X	X	152. Tiedown provisions damaged, missing, incorrectly located or not as specified.	3.19
X	-	X	X	153. Shipping data plate missing or not as specified.	3.20
X	-	X	X	154. Identification plates and marking missing or not as specified.	3.23

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TABLE I. Examination schedule. (Cont'd)

First Article(s) Pre-Test	First Article(s) Post Test	Initial Production	Quality Conformance	Defects	Requirement Paragraph(s)
1	2	3	4	5	6
X	-	X	X	155. Instruction plates missing or not as specified.	3.24
X	-	X	X	156. Treatment and painting not as specified.	3.25
X	-	X	X	157. Lubrication means not provided as specified or instructions not provided.	3.26
X	X	X	-	158. Maintenance not as specified.	3.27
X	-	X	X	159. Special tools not furnished (when required).	3.29
X	-	X	X	160. Toolbox damaged, incorrect size or not as specified.	3.30
X	-	X	X	161. Anti-vandalism protection damaged, missing or not as specified.	3.32
X	-	X	X	162. Maintenance forms case missing or not as specified.	3.33
X	-	X	X	163. Vehicle weight classification sign set missing or not as specified.	3.34
X	-	X	X	164. Workmanship not as specified.	3.35

4.4.1 Inspection failure. Failure of the inital production tractors to meet any requirements specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production tractors and shall be cause for refusal by the Government to continue acceptance of production tractors until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on tractors previously accepted and produced under the contract. Any deficiencies found as a result of the inital production inspection will be considered prima facie evidence that all tractors accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.5 Quality conformance inspection.

4.5.1 Examination. After successful completion of all tests specified in 4.5.2, each tractor shall be subjected to the examinations marked "X" in column 4 of table I. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each tractor shall be subjected to the tests marked "X" under column 3 of table II. Failure of any test shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with table I.

4.6.2 Tests.

4.6.2.1 Test conditions. Prior to test, the tractor shall be lubricated with oils and greases specified in 3.26. The oils shall be those designated for use in the ambient temperature at the place of test. The tractor shall be serviced, adjusted and operated for break-in as prescribed by the contractor. Only that maintenance established by the contractor and submitted as scheduled maintenace prior to commencement of the tests shall be performed during the tests. All lights shall be turned on during all operational tests. Prior to and throughout the testing the safety and health aspects of the tractor shall be evaluated and reported on in the test report.

4.6.2.2 Test schedule. Tests shall be in accordance with table II.

TABLE II. Test schedule.

First Article Test	Initial Production Test	Quality Conformance Test	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X	X	X	Test conditions.	4.6.2.1	3.5, 3.8, 3.26
-	-	X	Production run-in.	4.6.2.3	3.26
X	X	-	Maintainability.	4.6.2.4	3.27
X	X	-	Sound level.	4.6.2.5	3.5.1
X	X	-	Weight.	4.6.2.6	3.6.1
X	X	-	Dozing.	4.6.2.7	3.8.1
X	X	-	Ripping.	4.6.2.8	3.8.1
X	X	-	Ditching.	4.6.2.9	3.8.1
X	X	-	Winching.	4.6.2.10	3.8.1
X	X	-	Stability.	4.6.2.11	3.8.2
X	X	-	Fording.	4.6.2.12	3.8.3
X	X	-	Travel speeds.	4.6.2.13	3.8.4
X	X	-	Maximum drawbar pull.	4.6.2.14.1	3.8.5

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TABLE II. Test schedule. (Cont'd)

First Article Test	Initial Production Test	Quality Conformance Test	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X	X	-	Drawbar vs speed.	4.6.2.14.2	3.8.6
X	X	-	Brakes.	4.6.2.15	3.11.2
X	X	-	Front tow hook.	4.6.2.16	3.12.3
X	X	-	Electromagnetic radiation.	4.6.2.17	3.15.5
X	X	X	DCA capability.	4.6.2.18	3.15.8
X	X	-	DCA - STE/ICE interfaces.	4.6.2.18.1	3.18
X	X	-	Slinging provisions.	4.6.2.19	3.18
X	X	-	Tiedown provisons.	4.6.2.20	3.19

4.6.2.3 Production run-in. Start and operate each tractor for not less than 1/2 hour on a skid plate, or on a test track, to assure operation of all components and to allow for minor adjustments (see 3.26). All controls and attachments shall be manipulated and the tractor shall be operated in each gear selection in both forward and reverse directions. Evidence of contamination, leakage, permanent deformation, malfunction, or damage to the hydraulic system or other tractor components, or inability of any tractor control or component to operate as specified shall be cause for rejection of the tractor. The Government may require additional run-in after deficiencies are corrected. Any defect found during production run-in shall be prima facie evidence that tractors previously accepted are similarly defective unless evidence, satisfactory to the contracting officer, is furnished by the contractor that they are not similarly defective.

4.6.2.4 Maintainability. Prior to tests (see 4.6.2), perform the maintenance procedures as outlined in the maintenance schedule submitted by the contractor (see 3.27.3). Oil samples shall be taken from the engine and transmission. Inaccessability to lubrication fittings, drainage of fluid on the tractor, fluid leakage, inability to lubricate, incomplete listing in the maintenance schedule or inaccessability or nonconformance to requirements of 3.10.3 and 3.27 shall constitute failure of this test.

4.6.2.5 Sound level.

4.6.2.5.1 Interior sound level. The sound level at the operator's station shall be tested in accordance with SAE J919. The data shall include dB(A) and octave band pressure levels taken at the operator's station. Nonconformance to the requirements of 3.5.1 shall constitute failure of this test.

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4.6.2.5.2 Exterior sound level. The tractor shall be tested in accordance with SAE J88. Nonconformance to the requirement of 3.5.1 shall constitute failure of this test.

4.6.2.6 Weight. The tractor shall be weighed in accordance with 3.6.1. Inability to meet the requirements of 3.6.1 shall consitute failure of this test.

4.6.2.7 Dozing. Perform bulldozing for 300 hours by digging and backfilling a minimum of 36 rectangular pits in virgin or recompacted soil. Each pit shall be a minimum of 100 feet long, 50 feet wide and 4 feet deep. The pit shall be dug by moving the dirt in one direction and stockpiling it at the end of the pit. Inability of the tractor or dozer attachment to withstand 300 hours of operation without damage or permanent deformation shall constitute failure of this test.

4.6.2.8 Ripping. Using the two outboard ripper shanks, the type I tractor shall be operated ripping an area consisting of imbedded boulders in a cohesive soil having a minimum density of 95 percent of the maximum density as determined by the modified Proctor density test method. Ripping depth shall be not less than 24 inches. Using three shanks, the tractor shall be operated ripping an area consisting of a cohesive soil having a minimum density of 95 percent of the maximum density as determined by the modified Proctor density test method. Ripping depth shall be not less than 20 inches deep. Both the two shank and three shank ripping shall be performed for 50 hours each. Inability of the tractor and ripper attachment to accomplish 100 hours of operation without damage or permanent deformation, throwing of track or interference of track with the fender shall constitute failure of this test. This test is not required if type II tractor only is required by 6.2.

4.6.2.9 Ditching. Operate the tractor with dozer attachment ditching in virgin or recompacted soil, digging a V-ditch having side slopes of one vertical to three horizontal to a depth of not less than three feet and not less than 50 feet long. Perform the ditching for 100 hours; inability of the tractor or dozer attachments to withstand 100 hours of V-ditching without damage or permanent deformation shall constitute failure of this test.

4.6.2.10 Winching. Perform winching operation with the type II tractor for 2 hours by intermittently applying a bare drum line pull of not less than 50,000 pounds at a line speed of not less than 80 feet per minute. This test is not required if type I tractor only is required by 6.2. Any malfunction or damage of the tractor or winch (except cable breakage) shall constitute failure of this test.

4.6.2.11 Stability. The type II tractor shall be tested for slope operation in accordance with SAE J897. Prior to the static portion of the test, the engine oil level shall be adjusted to the low mark on the oil level indicator. During the static test, the tractor shall be parked on a 35 percent slope. The engine shall run at idle speed for five minutes during each tractor orientation required. Prior to the dynamic test, the engine oil level shall be adjusted to the full mark on the oil level indicator. During the dynamic test, the tractor

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shall be operated for one hour on a 60 percent slope of at least 50 feet in length for each orientation of the tractor centerline parallel with the slope. The tractor shall be operated for one hour on a 35 percent slope of at least 50 feet in length when the tractor centerline is perpendicular with the slope. Nonconformance to 3.8.2, evidence of leaks, lack of lubrication brake overheating or seizing, or inability of the tractor engine to maintain constant oil pressure within the operating range on 35 percent side slopes and 60 percent longitudinal slopes with the engine lubricating oil at the low level and then at the high level shall constitute failure of this test. The test is not required if the type I tractor only is required by 6.2.

4.6.2.12 Fording. The tractor shall be driven into a body of fresh water to a depth of not less than 30 inches and large enough to allow operation of the tractor. The tractor shall be operated in low gear forward and reverse for a period of 15 minutes in 30 inches of water, then driven on land in high gear for 15 minutes, then returned to the water. The engine shall be stopped for one minute. This cycle shall be performed five times. All reservoirs and gear and bearing enclosures shall be sampled for presence of water prior to and after the test. Evidence of water leakage, damage or malfunction of any component shall constitute failure of this test.

4.6.2.13 Travel speeds. Test the tractor on a level, dry surface of sufficient length to attain maximum speed prior to entering the measured test course. Operate the tractor at not more than full load governed speed between the initial and final points that are not less than 100 feet apart. Measure the lapsed time that is required to cover the measured distance. Repeat for a total of three times in each direction for each gear forward and reverse. Average speeds less than those required in 3.8.4 shall constitute failure of this test.

4.6.2.14 Drawbar pull.

4.6.2.14.1 Maximum drawbar pull. The tractor shall be tested for maximum drawbar pull in accordance with SAE J872. A drawbar pull of less than 35,000 pounds at 1.4 mph shall constitute failure of this test.

4.6.2.14.2 Drawbar pull vs speed. Tractor shall be tested in accordance with SAE J872 to include the approximate ground speed in each gear as follows:

a. Twenty-five percent of maximum ground speed.
b. Fourth-five percent of maximum ground speed.
c. Sixty percent of maximum ground speed.
d. Eighty percent of maximum ground speed.
e. Ninety percent of maximum ground speed.

A drawbar pull vs speed curve shall be plotted and compared to figure 1. Inability to equal the curve in figure 1 shall constitute failure of this test.

4.6.2.15 Brakes. Both the steering and parking brakes shall be tested by applying a drag force on the tow hook, then the drawbar. The tractor shall be parked on a level surface that will allow full grouser penetration with minimum track shoe penetration. Inability of the brakes to prevent track rolling when a 35,000 pound drag force is applied shall constitute failure of this test.

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4.6.2.16 Front tow hook. The tow hook shall be tested by restraining the tractor and applying a horizontal load of not less than 75,000 pounds to the tow hook. Inability to withstand this force, breakage, permanent deformation of the hook or its mounting or nonconformance to the requirements of 3.12.3 shall constitute failure of this test.

4.6.2.17 Electromagnetic radiation. The tractor shall be tested in accordance with SAE J551. Measured levels in excess of the recommended levels shall constitute failure of this test.

4.6.2.18 DCA capability. The contractor will perform a 100 percent visual inspection of all DCA wiring harnesses prior to installation on the tractor to insure wire and connections are free of error. After installation on the tractor, test the DCA harness (reference chapter 5 of design guide for DCA) on each vehicle with a DCA tester prior to interfacing with STE/ICE. Nonconformance to 3.15.8 shall constitute failure of the DCA.

4.6.2.18.1 DCA - STE/ICE interfaces. Upon completion of inspection and tests specified in 4.6.2.18, test the DCA for all measurements specified in 3.15.8 using STE/ICE. Nonconformance to 3.15.8 shall constitute failure of DCA.

4.6.2.19 Slinging provisions. The slinging provisions shall be demonstrated to prove conformance to 3.18. Nonconformance to 3.18, weld failure, or permanent deformation of the tractor or the slinging provisions shall constitute failure of this demonstration. The demonstration procedure shall be as follows:

a. Lift the tractor and hold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the tractor. Spreader bars are permitted. Determine the force and angles of application for each provision. Measure the clearance between each sling and the tractor.
b. Restrain the tractor by anchoring the main frame and subject each slinging provision to a force equal to the working load specified in MIL-STD-209. This force shall be applied in the direction as determined in "a" above. Hold each load for a minimum of 90 seconds.

4.6.2.20 Tiedown provisions. The tiedown provisions shall be demonstrated to prove conformance to 3.19. Apply a force equal to the limit load as specified in MIL-STD-209 to each of the tiedown provisions. Weld failure or permanent deformation of the tractor or the tiedown provisions shall constitute failure of this demonstration.

4.6.2.21 Certifications. The contractor shall certify in writing to the contracting officer that the tractor complies with the safety, OSHA and Federal laws requirements of 3.1, the environmental conditions of 3.7, the composition II requirements of 3.7.2 (if required), the components requirements of 3.9, the engine requirements of 3.10, the horsepower requirement of 3.10.1, the winch hook capacity of 3.16.2, the transportability requirements of 3.17, the air transportability requirements of 3.17.1 and, if applicable, the heater capacity of 3.5.3 and 3.5.4. If acceptable certification is not supplied, these requirements shall be tested. Nonconformance shall constitute failure of these tests.

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4.6.2.22 Certified test reports. The contractor shall furnish certified test reports to prove compliance with the ROPS/FOPS requirements of 3.5.2 and, if applicable, the compliance with the cab requirements of 3.5.3 and 3.5.4. If certified test reports are not available, these requirements shall be tested. Nonconformance to these requirements shall constitute failure of these tests.

4.7 Inspection comparison. The Government may select tractors at any time during the contract production period and subject these tractors to the examination and tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production tractors. In addition to any test specified as part of the inspection comparision, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparision.

4.7.1 Inspection failure. Failure of an inspection comparison tractor to meet any requirement specified herein during and as a result of the examination and tests specifed in 4.7 shall be cause for rejection of the inspection comparison tractor(s) and shall be cause for refusal by the Government to continue acceptance of production tractors until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiences shall be accomplished by the contractor at no cost to the Government on tractors previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all tractors accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging. The preservation, packing, and marking shall be examined to determine compliance with the quality assurance provisions of MIL-T-3351 and the specific requirements specified in section 5 of this document.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each complete tractor, components and attachments shall be preserved in accordance with the level A preservation requirements of MIL-T-3351 except engine cooling systems shall be preserved by the water and antifreeze procedure specified in MIL-E-10062, using antifreeze conforming to MIL-A-46153.

5.1.2 Level B. Each complete tractor, components, and attachments shall be preserved in accordance with the level B preservation requirements of MIL-T-3351 except engine cooling systems shall be preserved by the water and antifreeze procedure specified in MIL-E-10062, using antifreeze conforming to MIL-A-46153.

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5.1.3 Level C. Each complete tractor and all items issued with the tractor shall be suitably preserved for retail distribution directly to a using customer. In addition, the following specific requirements shall be accomplished: Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coatings, volatile corrosion inhibitors, or desiccated packs. Items requiring protection from physical damage to surfaces, or which are fragile by nature, shall be protected by wrapping, pack compartmentization, or cartonizing of the individual item. The cooling system shall be filled with a fresh solution of 50 percent water and 50 percent antifreeze, conforming to MIL-A-46153, and a tag attached indicating the temperature to which the cooling system can be subjected to before freezing. The crankcase shall contain the required amount of lubricating oil. A guide shall be furnished indicating the preservation and packing materials to be removed, in detail, and any other servicing required prior to placing the tractor in operation.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Each complete tractor, preserved as specified in 5.1, shall be packed in accordance with the level A packing requirements of MIL-T-3351, as specified for type II unboxed (mobile) tractors.

5.2.2 Level B. Each complete tractor, preserved as specified in 5.1, shall be packed in accordance with level B packing requirements of MIL-T-3351, as specified for type II unboxed (mobile) tractors.

5.2.3 Level C. Each complete tractor preserved as specified in 5.1, shall be prepared for mobile shipment in a manner to assure carrier acceptance and safe delivery to destination at the lowest ratings of the carrier.

5.3 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tractors are intended for dozing, ditching, ripping, excavation, backfilling, winching, towing and for push loading scrapers.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type tractors required (see 1.2).
c. Date of issue of the DoDISS applicable to this contract and exceptions thereto (see 2.1.1).
d. Number and type of preproduction model tractors, time frame required for submission, and whether the Government will conduct any or all of the examinations and tests. When the Government will conduct some but not all of the preproduction examiantions and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).

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e. When initial production testing is required and number and type of initial production tractors to be furnished (see 3.3).
f. When a ROPS/FOPS enclosed cab is required (see 3.5.3).
g. When winterized cab is required and number and type of tractors (see 3.5.4).
h. Time frame for submitting vehicle classification data sheets (see 3.21).
i. Technical publications required (see 3.28).
j. Time frame for submission of technical information package (see 3.31).
k. When GFE is required to be furnished (see 3.33).
l. Time frame for submitting first article test report (see 4.3.3).
m. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Government-furnished equipment. The contracting officer should arrange to furnish the equipment specified in 3.33.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 2410-A042

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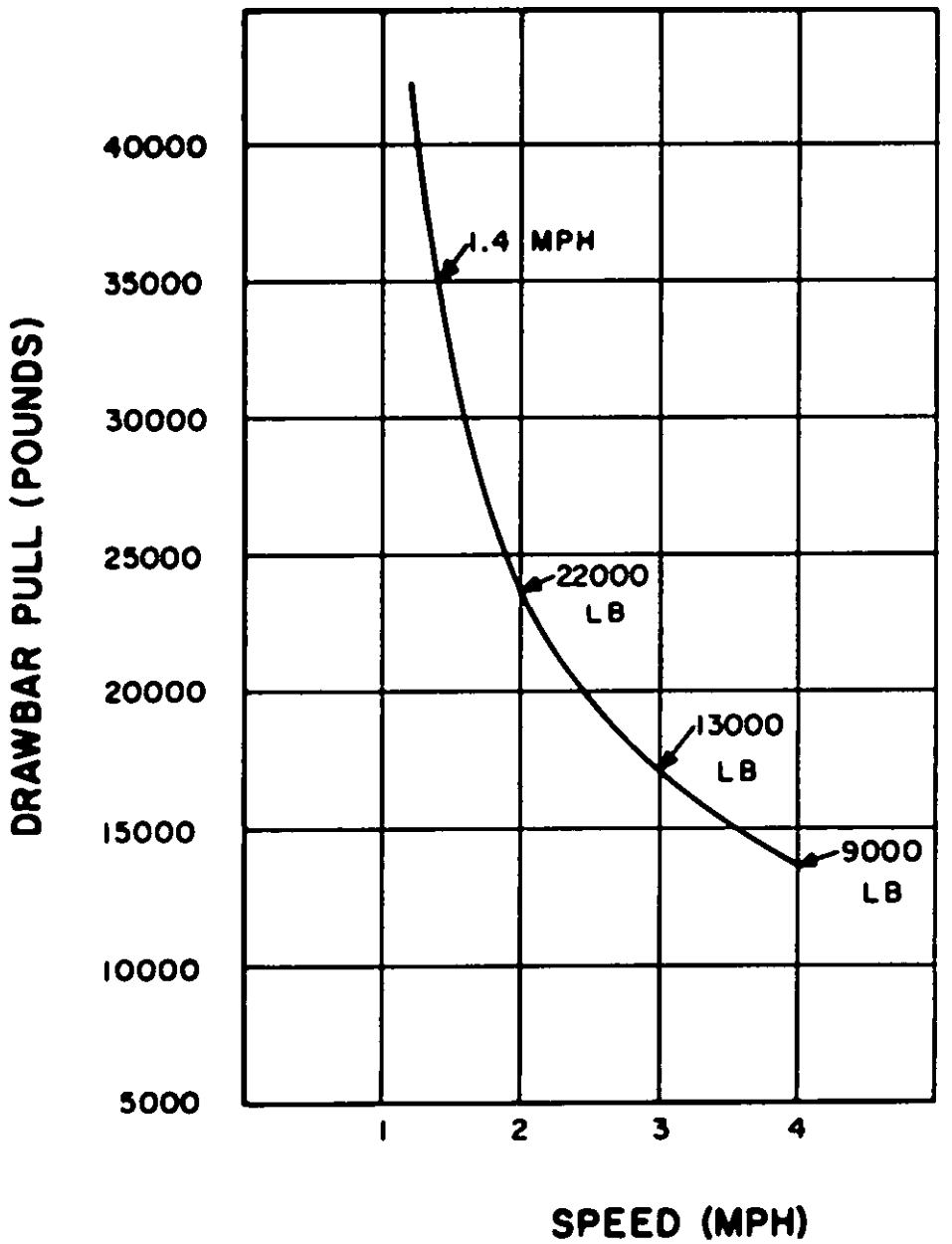


FIGURE 1. Drawbar pull vs speed curve.

X-601A

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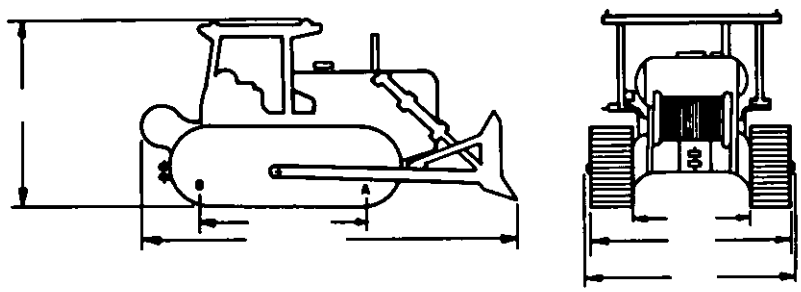
VEHICLE CLASSIFICATION DATA
TRACTOR, FULL-TRACKED
DATA REQUIRED FOR IDENTIFICATION AND LOAD CLASSIFICATION OF SUBJECT VEHICLE

Load Class		
E	C	H

NOMENCLATURE:

Mfr.	Model No.	
NSN	Agency	Stock No.

Traveling Load Category		Total Weight W	Pay Load	Max. Towed Load	Max. Pintle Load
Empty	E		X		
Cross Country or Off Highway	C				
Laden or On Highway	H				



Load			
Traveling Load Category	A–B	psi A–B	Pintle Max.
Empty E			
Cross Country On or Off Highway C			
Laden or On Highway H			

All loads in short tons.
All loads to include weight of crew and appurtenances.
Class No's. for vehicles over 3,000 lbs.
Longitudinal dimensions in feet.
Vertical and transverse dimensions in inches.

FIGURE 2.

X-3190

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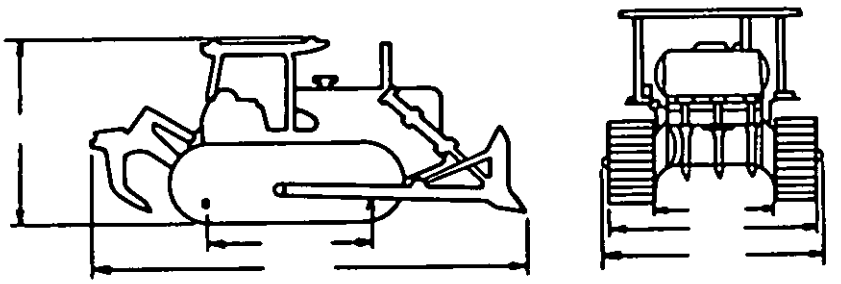
VEHICLE CLASSIFICATION DATA
TRACTOR, FULL-TRACKED
DATA REQUIRED FOR IDENTIFICATION AND LOAD CLASSIFICATION OF SUBJECT VEHICLE

Load Class		
E	C	H

NOMENCLATURE:

Mfr.	Model No.	
NSN	Agency	Stock No.

Traveling Load Category		Total Weight W	Pay Load	Max. Towed Load	Max. Pintle Load
Empty	E		X		
Cross Country or Off Highway	C				
Laden or On Highway	H				



Load			
Traveling Load Category	A–B	psi A–B	Pintle Max.
Empty E			
Cross Country On or Off Highway C			
Laden or On Highway H			

All loads in short tons.
All loads to include weight of crew and appurtenances.
Class No's. for vehicles over 3,000 lbs.
Longitudinal dimensions in feet.
Vertical and transverse dimensions in inches.

FIGURE 3 .

X-3189

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-0052270D(ME)	Tractor, Full-Tracked, Low-Speed: Diesel-Engine-Driven-Medium-Drawbar Pull

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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